



M2Unity Pipeline Suite v1.0.2

Professional Documentation

Maya-to-Unity static mesh preflight, cleanup, collider proxy validation, role naming, reporting and FBX export toolkit.

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Public documentation package
Crafted with care over a few cups of coffee.

Documentation scope

This PDF combines the README, the full feature reference, the licence/terms document and the provided UI screenshots into one detailed user-facing manual. It avoids private machine paths and uses only generic example paths where a path is needed.

Prepared for artists, technical artists, asset creators and pipeline users who move static mesh assets from Autodesk Maya to Unity.

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1. Executive Overview

M2Unity Pipeline Suite is a Maya-side preparation and export toolkit for static mesh assets intended for Unity. It is not just an FBX exporter. It is a structured preflight assistant that checks scene readiness, applies optional safe cleanup, manages role-based naming, matches collider and trigger proxy meshes, exports FBX files and writes QA reports.

The main goal is to catch common Maya-to-Unity asset problems before they become import, prefab or collider setup problems inside Unity. The workflow is designed to be repeatable, artist-readable and safe enough for day-to-day asset preparation.

Core production rule

Mesh_ is used for visible render meshes. COL_Mesh_ is used for collider proxy meshes. TRG_Mesh_ is used for trigger proxy meshes. Safe Fix is for cleanup; role assignment is handled separately by dedicated utility buttons.

Who this tool is for

- 3D artists preparing static mesh assets in Autodesk Maya for Unity.
- Technical artists who need consistent naming, validation and export rules.
- Asset creators who need a clear package workflow for props, modular pieces and environment kits.
- Pipeline users who want JSON/HTML reports for review, debugging and handoff.

What v1.0.2 focuses on

- Role-based naming for render meshes, collider proxies and trigger proxies.
- Maya import / scene prefix cleanup before standardized names.
- Safe Fix cleanup that does not accidentally convert render meshes into colliders.
- Export-time refresh of matched COL_ and TRG_ proxy objects.
- Generated Unity AssetPostprocessor C# helper for automated import handling.

2. Visual Overview

The following screenshots show the main interface, rule tabs, role naming workflow, analyze/export behavior and generated report output. Captions explain where each screen fits in the production workflow.

The Dashboard / Wizard tab is the central workflow area. The user chooses a preset, analyzes selected assets, applies safe fixes, revalidates, and exports FBX files plus reports.

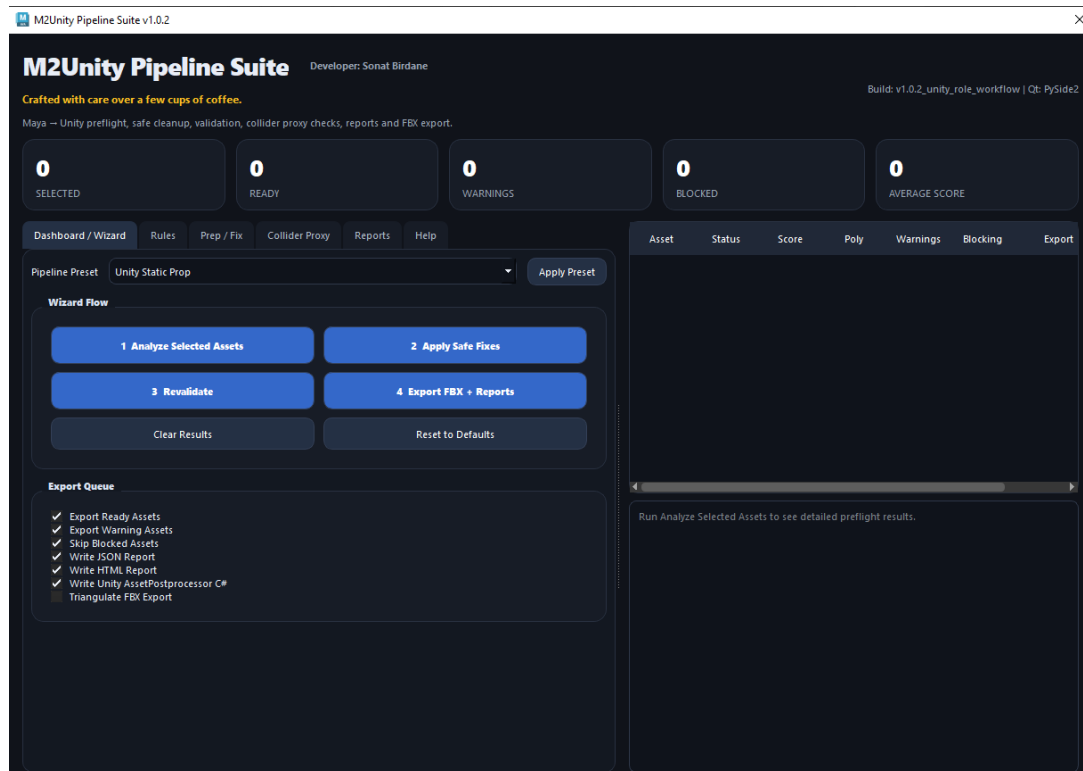


Figure 1 - Dashboard / Wizard tab with preset selector, main workflow buttons, export queue and status cards.

Rules begin with scope and naming behavior. This is where render mesh prefix validation, scene prefix cleanup and selected mesh analysis rules are controlled.

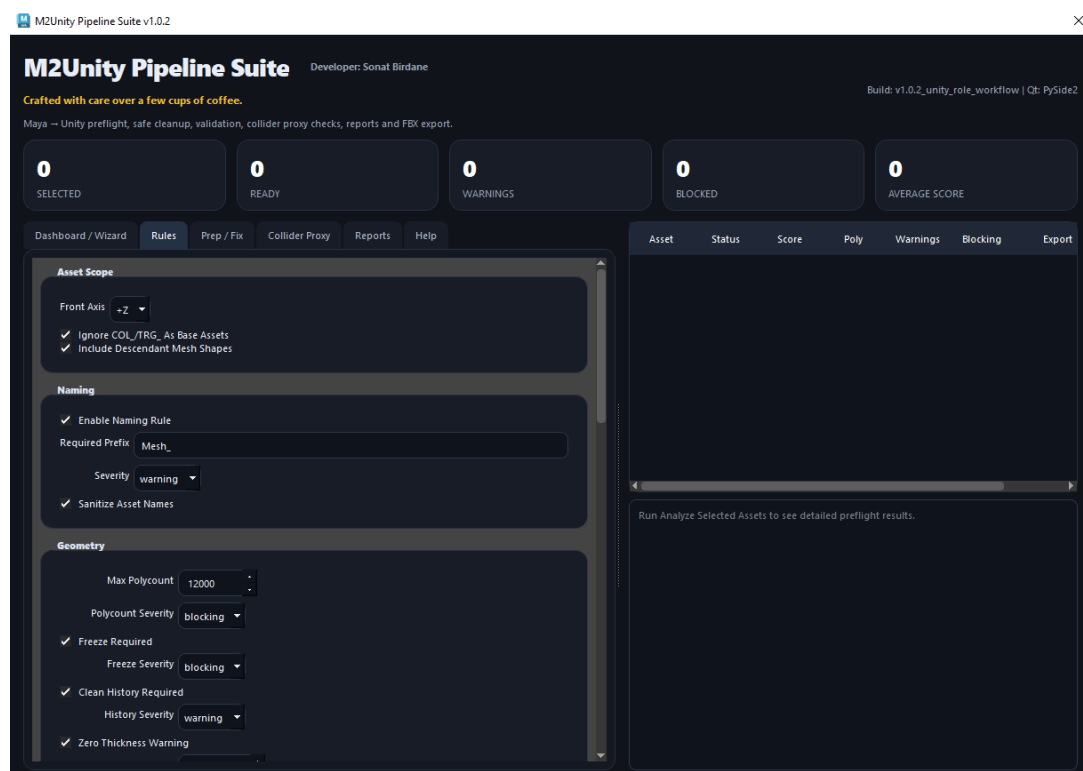


Figure 2 - Rules tab: asset scope and naming options.

Geometry rules cover polycount, freeze requirement, construction history, zero-thickness checks and dimension validation. Severity can be warning or blocking.

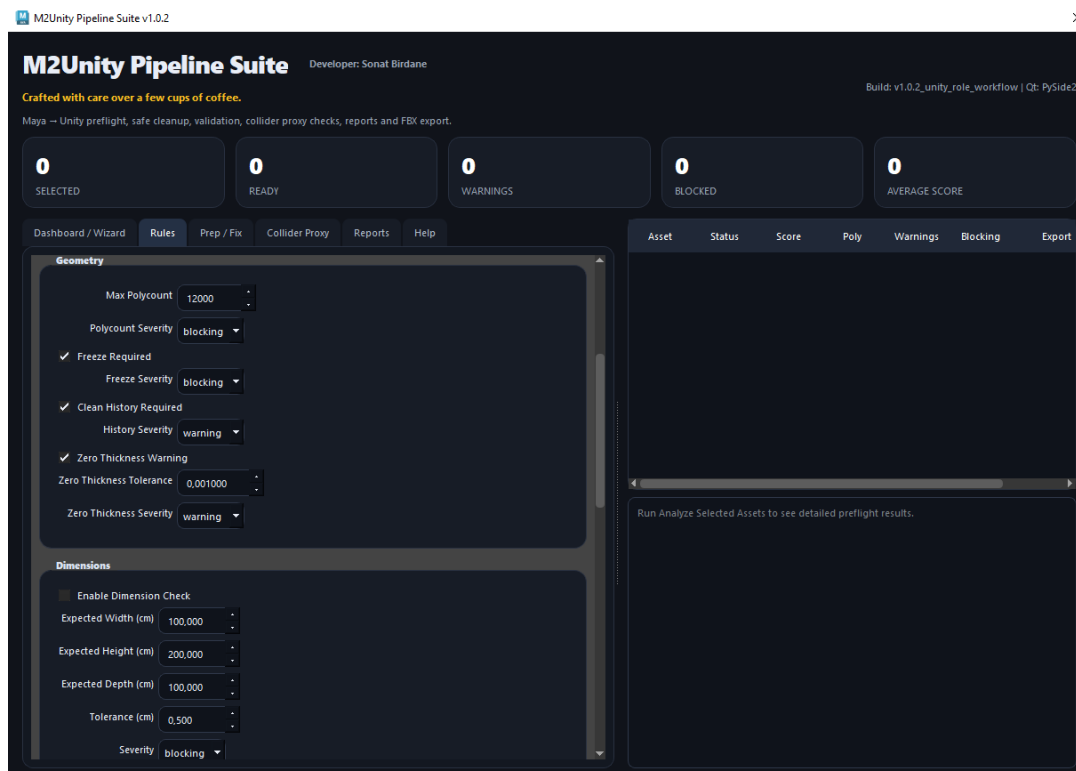


Figure 3 - Rules tab: geometry and dimension validation.

Pivot and grid checks keep assets modular and predictable in Unity. Advanced readiness checks include LOD, material, UV and socket-style helper rules.

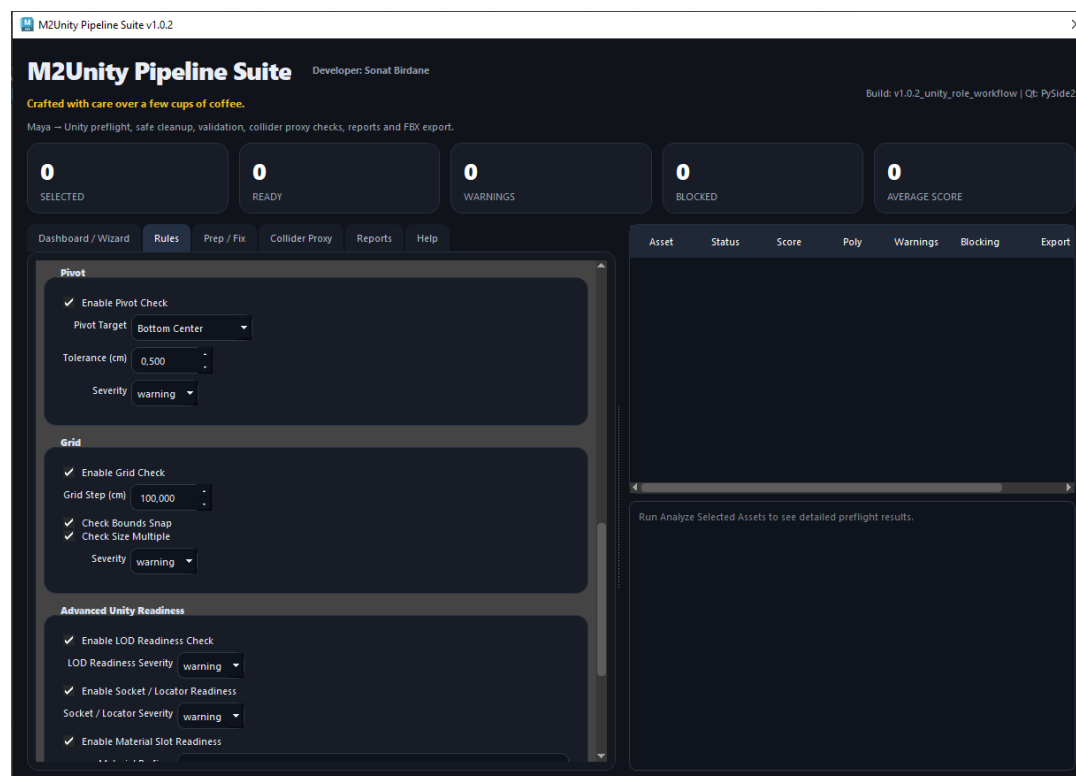


Figure 4 - Rules tab: pivot, grid and advanced Unity readiness checks.

The advanced rule area helps users enforce Unity-friendly static mesh practices, such as material slot limits, UV set readiness and lightmap UV expectations.

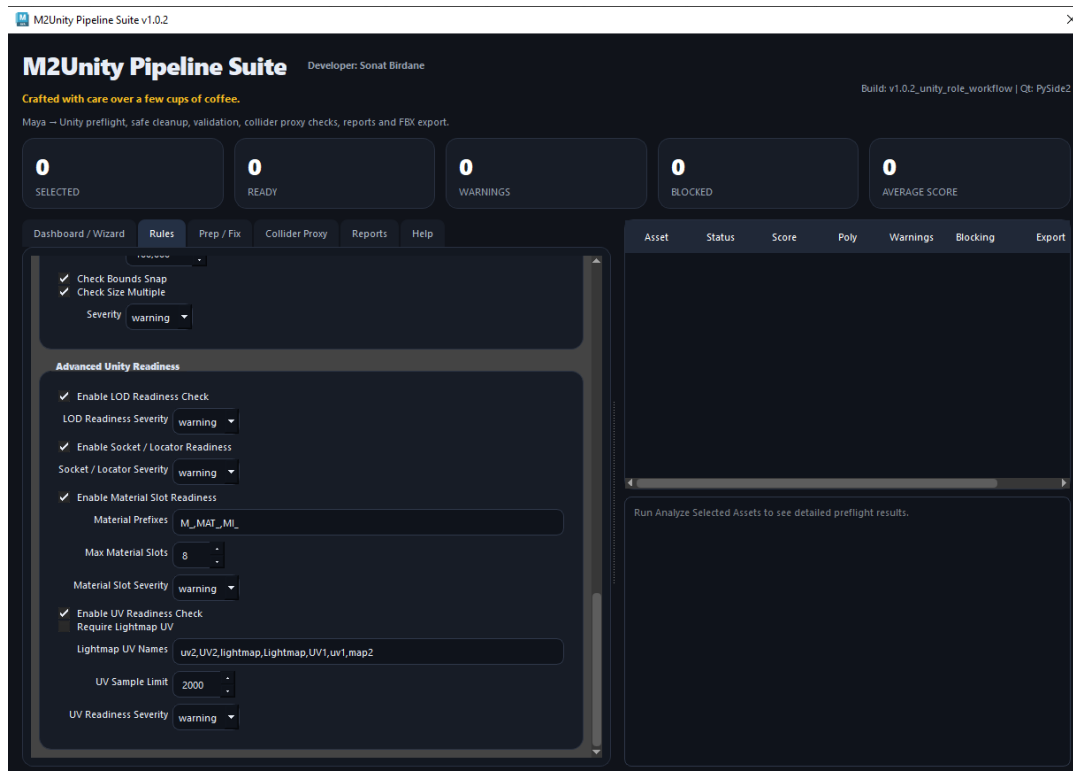


Figure 5 - Advanced readiness configuration for LOD, material and UV checks.

Safe Fix can duplicate backups, hide backups, rename render meshes, freeze transforms, delete history, move pivots and create helper previews. It does not resize or rebuild geometry automatically.

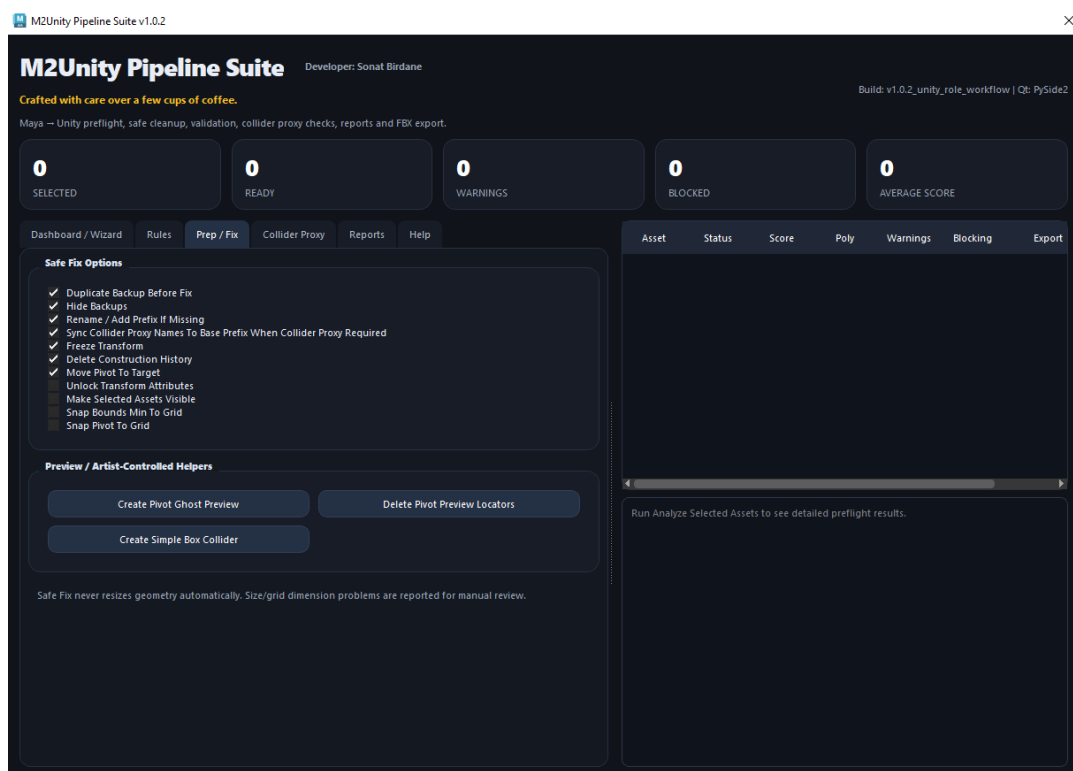


Figure 6 - Prep / Fix tab with safe cleanup options and preview helpers.

The Collider Proxy tab is where Unity-specific COL_ and TRG_ proxy workflows are managed. Role utility buttons can convert selected objects into render, collider or trigger roles.

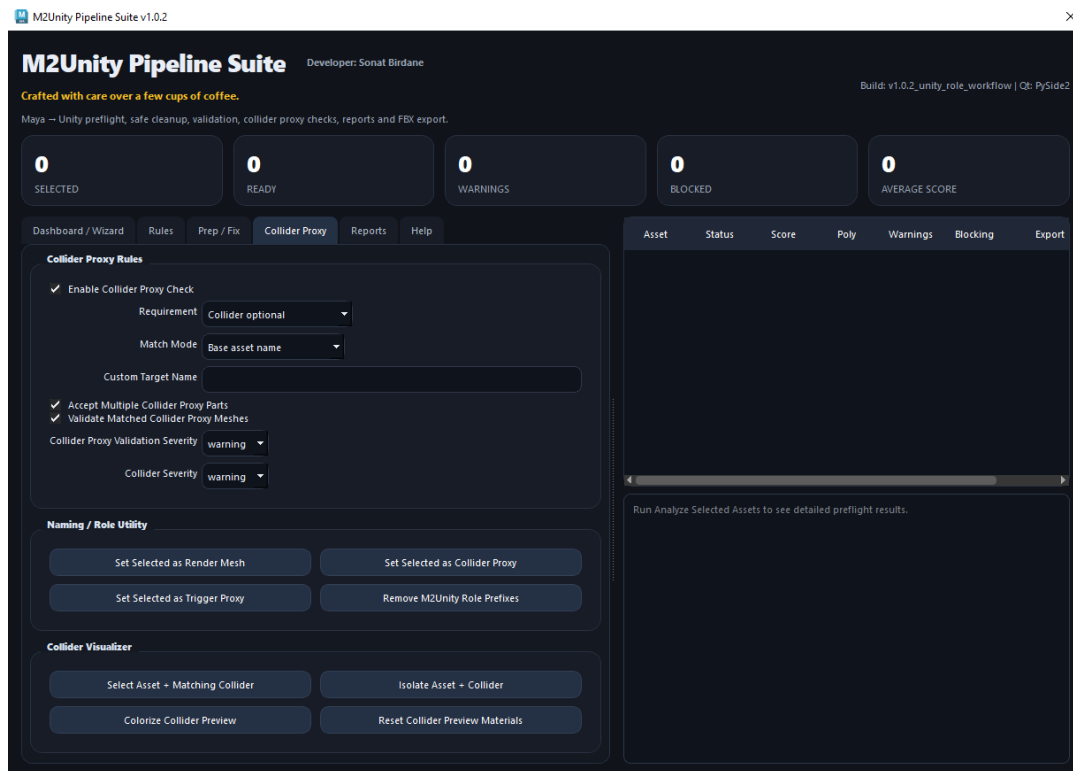


Figure 7 - Collider Proxy tab with Collider Rules, Naming / Role Utility and visualizer tools.

The Help tab summarizes the safest order of operations and reminds the user that generated Unity C# scripts are copied into Unity projects, not executed inside Maya.

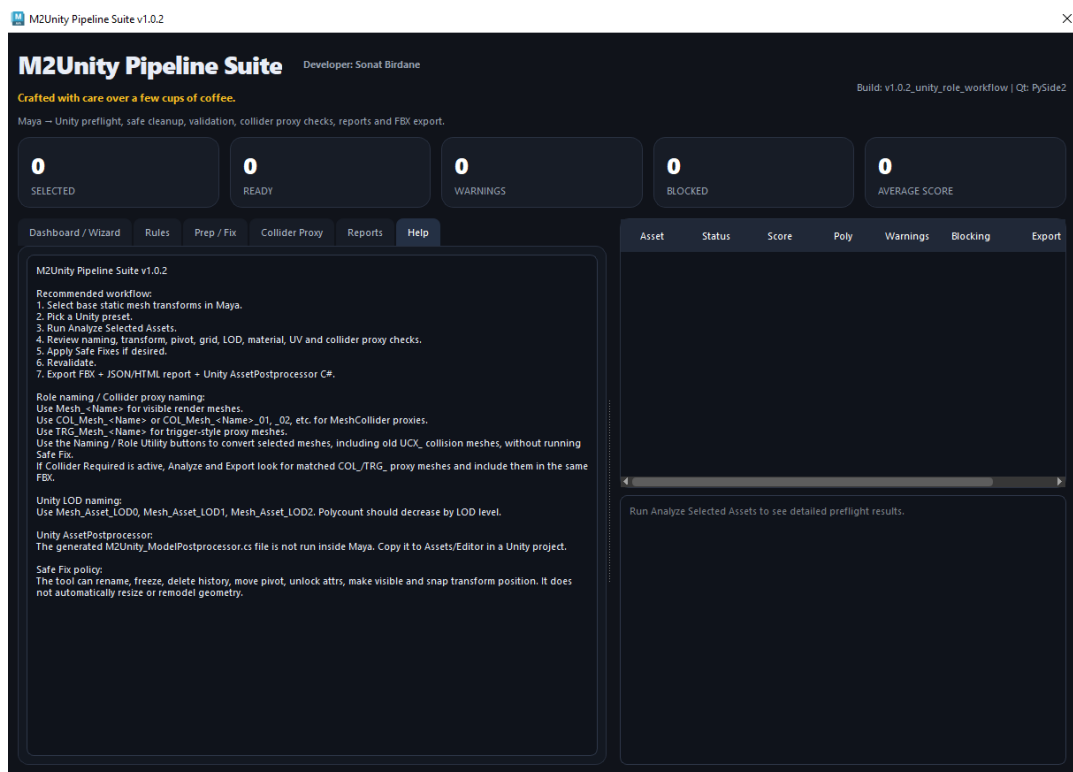


Figure 8 - Help tab with recommended workflow and role naming summary.

3. Core Concepts

Five-stage workflow

Stage	Purpose	Scene modification
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Analyze	Inspects selected Maya render mesh transforms and creates validation results.	No
Role Naming	Assigns selected objects to Mesh_, COL_Mesh_ or TRG_Mesh_ roles.	Yes - renames selected transforms
Safe Fix	Applies optional cleanup such as backup, rename, freeze, history delete, pivot move and grid snapping.	Yes - controlled by options
Revalidate	Runs analysis again after naming or cleanup changes.	No direct modification
Export	Exports FBX files, matched COL_/TRG_ proxies, reports and optional Unity C# helper.	Writes files to disk

Status levels

Status	Meaning	Default export behavior
Ready	No blocking issues were found.	Exported when Export Ready Assets is enabled.
Warning	Only warnings were found.	Exported when Export Warning Assets is enabled.
Blocked	One or more blocking issues were found.	Skipped when Skip Blocked Assets is enabled.

Severity system

Many validation rules use the same severity choices: Off, Warning and Blocking. Off disables the issue path, Warning reports the problem without necessarily blocking export, and Blocking prevents production export when Skip Blocked Assets is enabled.

M2Unity Score

Each analyzed asset receives a M2Unity Score. The score starts from a clean state and is reduced by warnings, blocking issues and failed validation checks. It is intended as a quick readability signal; it does not replace manual QA.

90-100 Very clean
70-89 Usable but review warnings
40-69 Risky or incomplete
0-39 Not production-ready without fixes

4. Supported Environment, Package Contents and Installation

Supported software

- Target Maya version: Autodesk Maya 2024 and newer.
- Required Maya components: Python 3, maya.cmds and the Maya FBX plug-in for FBX export.
- Qt compatibility: PySide2 and PySide6 fallback handling.
- Recommended Unity usage: a Unity project that imports FBX model assets and allows an Assets/Editor folder when using the generated AssetPostprocessor C# file.

Package contents

File	Purpose
m2unity_pipeline_suite_v1_0.py	Main Maya tool.

M2Unity Pipeline Suite.py	Simple launcher script for Maya.
M2Unity_Pipeline_Suite_v1_0_Easy_Installer.py	Copies tool files and creates/updates the Maya shelf button.
reinstall_m2unity_pipeline_suite_shelf_button.py	Recreates the shelf button without reinstalling all files.
m2unity_pipeline_suite_icon.png	Shelf button icon.
README / FUNCTIONS / LICENCE TXT files	Public release documentation, detailed function reference and licence terms.

Installation steps

- Extract the public release package to a temporary folder.
- Open Autodesk Maya.
- Open Windows > General Editors > Script Editor.
- Switch to the Python tab.
- Open or drag M2Unity_Pipeline_Suite_v1_0_Easy_Installer.py into the Python tab.
- Run the script and choose an install folder.
- The installer copies required files and creates or updates the M2U_Tools shelf.
- After installation, launch the tool from Maya Shelf > M2U_Tools > M2Unity Pipeline Suite.

Example-only install folder:
C:/Example/MayaTools/M2Unity_Pipeline_Suite

Example-only Unity postprocessor path:
D:/ExampleUnityProject/Assets/Editor/M2Unity_ModelPostprocessor.cs

Important

The generated Unity C# script is not executed inside Maya. It is intended to be reviewed and placed in a Unity project under an Editor folder.

5. Quick Start Workflow

The recommended workflow keeps role assignment, safe cleanup and export validation separate. This is important because collider and trigger role conversion should always be intentional.

Step	Action	Expected result
1	Select visible static render meshes in Maya.	The tool has a clear render mesh selection to analyze.
2	Use Set Selected as Render Mesh if needed.	Names become Mesh_<AssetName>.
3	Select collider or trigger proxy meshes and use role utility buttons if needed.	Names become COL_Mesh_<AssetName> or TRG_Mesh_<AssetName>.
4	Choose a preset and click Analyze Selected Assets.	The dashboard, result table and detail panel update.
5	Apply Safe Fixes only when cleanup is desired.	Backups, rename, freeze, history delete and pivot fixes run according to options.
6	Revalidate.	The current scene state is checked again.

7	Export FBX + Reports.	Eligible assets export with matched proxy meshes and reports.
8	Copy M2Unity_ModelPostprocessor.cs to Unity if desired.	Unity can convert COL/TRG_ objects into MeshColliders during import.

After Analyze, the tool displays total selected assets, readiness status, warning/blocking counts and detailed issue text for the highlighted asset.

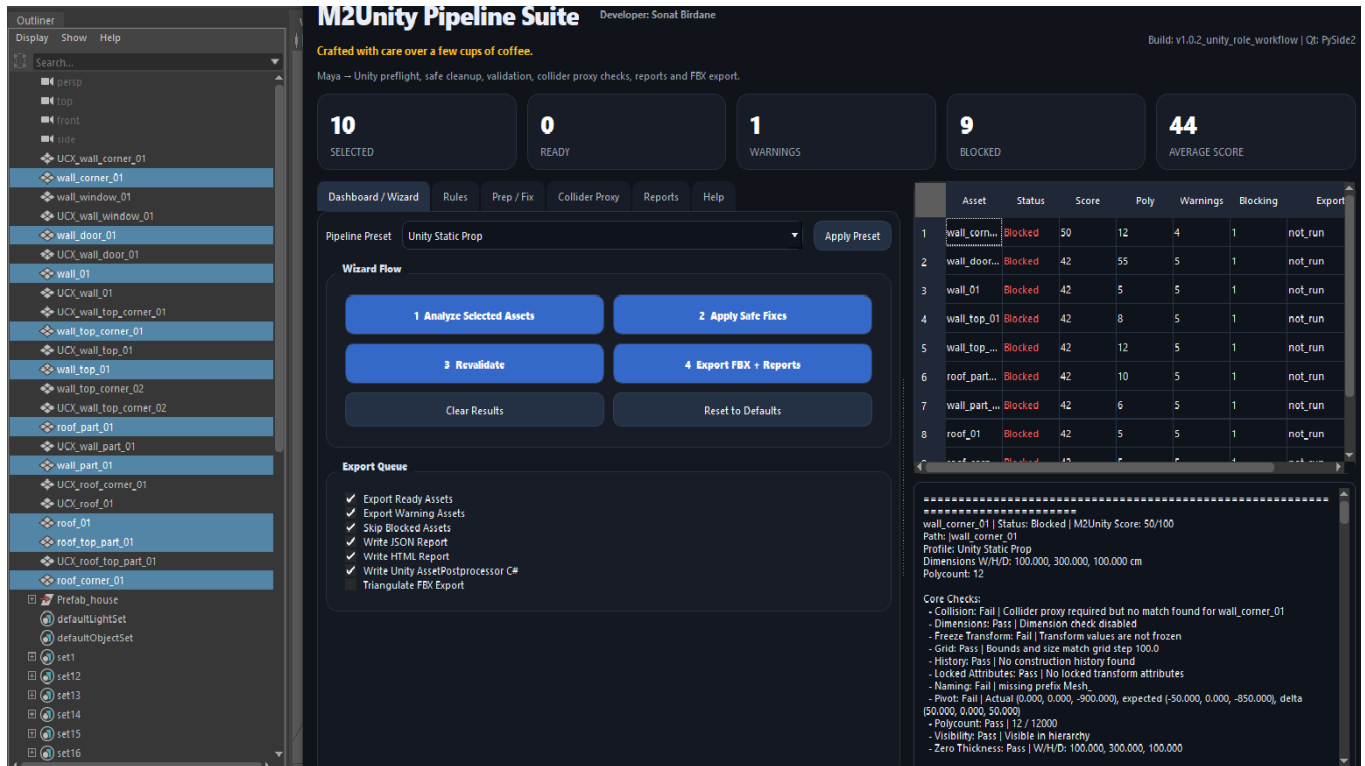


Figure 9 - Analyze result with Maya Outliner, status cards, result table and detail panel.

The detail panel makes issues readable: collider requirement failures, naming problems, transform warnings and fix recommendations are grouped per asset.

	Asset	Status	Score	Poly	Warnings	Blocking	Export
1	wall_corn...	Blocked	50	12	4	1	not_run
2	wall_door...	Blocked	42	55	5	1	not_run
3	wall_01	Blocked	42	5	5	1	not_run
4	wall_top_01	Blocked	42	8	5	1	not_run
5	wall_top_...	Blocked	42	12	5	1	not_run
6	roof_part...	Blocked	42	10	5	1	not_run
7	wall_part_...	Blocked	42	6	5	1	not_run
8	roof_01	Blocked	42	5	5	1	not_run

- Sockets: Pass No sockets found. This is valid for simple static meshes.							
Collider Proxy:							
Target: wall_corner_01							
- No matching collider proxy mesh							
Blocking Issues:							
- Freeze Transform requirement not satisfied							
Warnings:							
- Naming issue: missing prefix Mesh_							
- Pivot does not match target: Bottom Center							
- Missing required collider proxy for target: wall_corner_01							

Figure 10 - Detail panel with blocked assets, warnings and issue breakdown.

The workflow supports selecting multiple scene objects, applying role naming utilities, then re-running Analyze to verify the updated names and proxy matches.

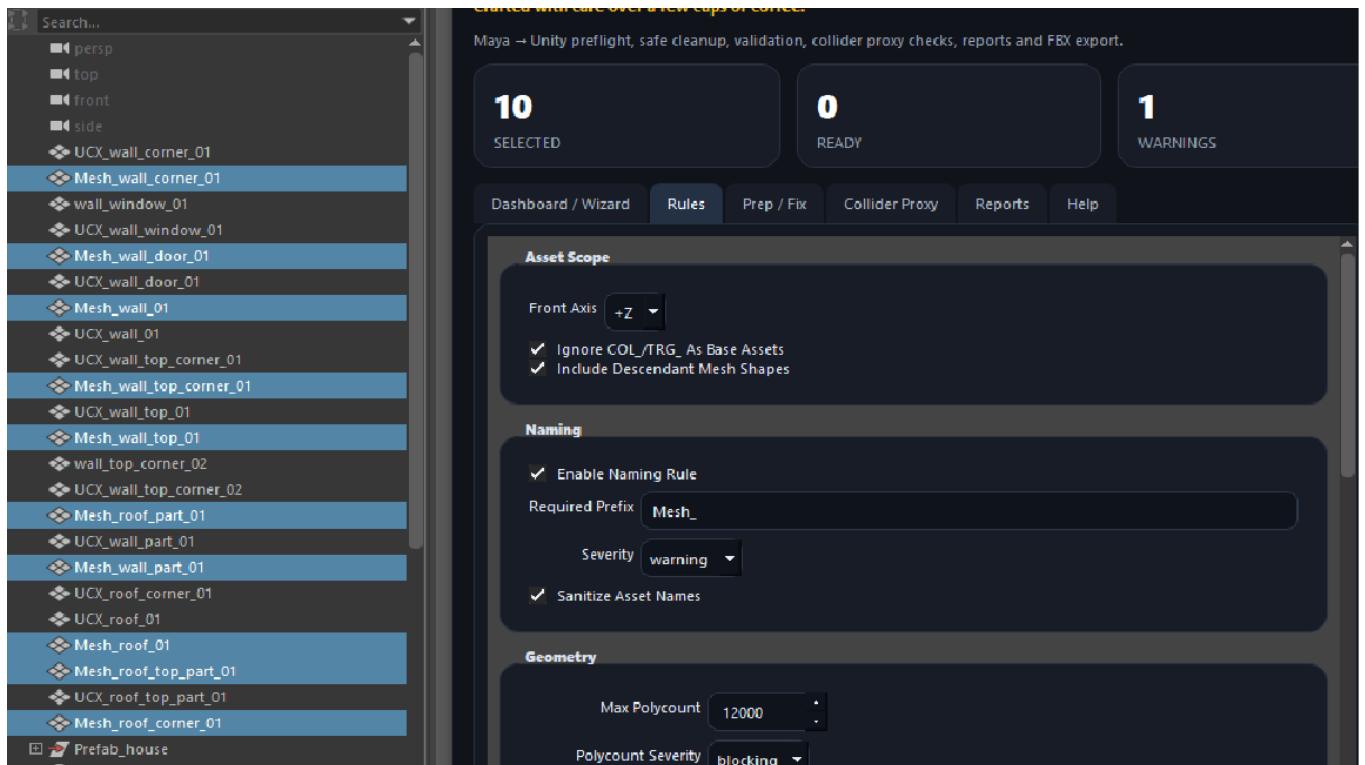


Figure 11 - Scene selection and rule review during role/naming preparation.

When Collider Required is active, matching COL_ or TRG_ proxies must exist. Role buttons remain separate from Safe Fix so the user explicitly controls role conversion.

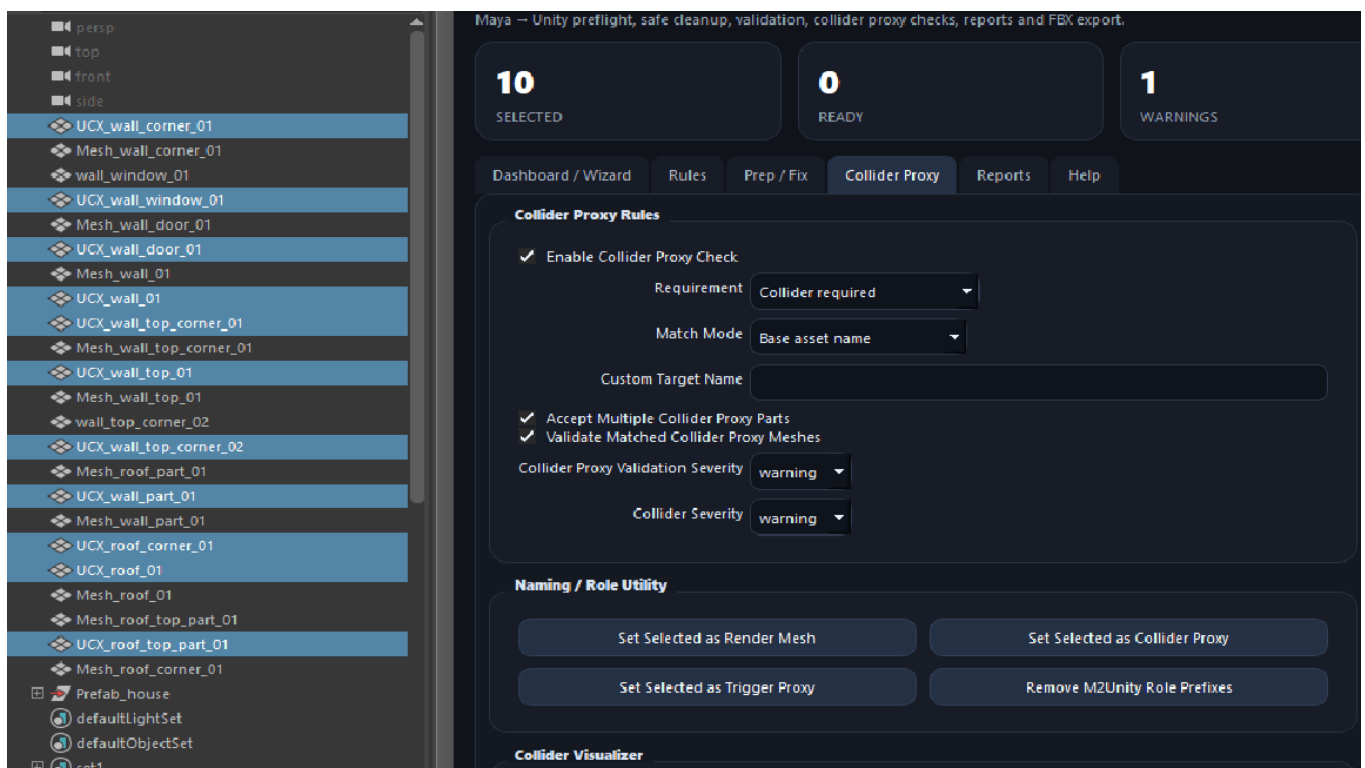


Figure 12 - Collider Required mode and role utility buttons.

This screenshot shows the conversion confirmation dialog. Unreal-style UCX objects can be renamed into Unity-oriented COL_Mesh_proxy names for matching/export.

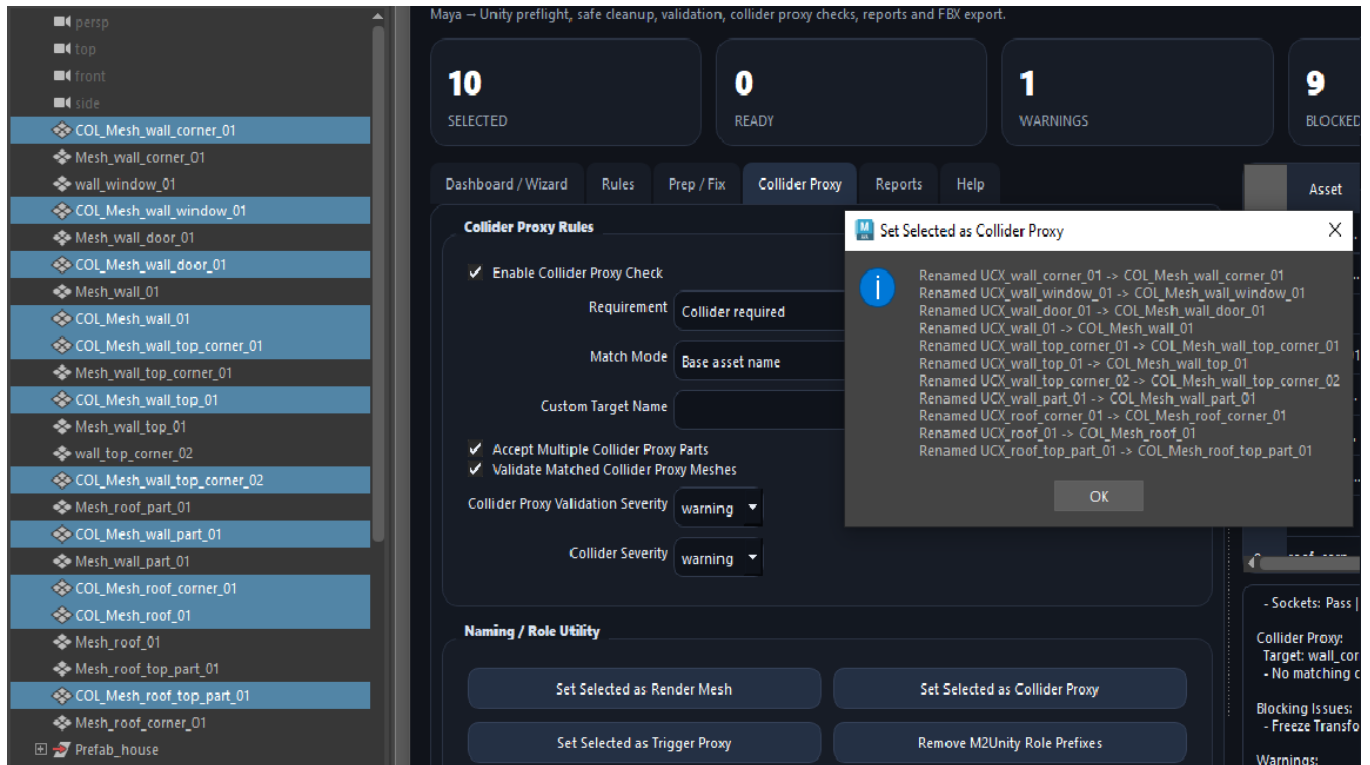


Figure 13 - Old UCX objects converted to COL_Mesh_ collider proxies.

The export result includes FBX files, M2Unity_Pipeline_Report.html, M2Unity_Pipeline_Report.json and M2Unity_ModelPostprocessor.cs.

M2Unity_ModelPostprocessor.cs	16.05.2026 18:28	CS Dosyası	3 KB
M2Unity_Pipeline_Report	16.05.2026 18:28	Firefox HTML Doc...	63 KB
M2Unity_Pipeline_Report	16.05.2026 18:28	JSON Dosyası	122 KB
Mesh_roof_01_001	16.05.2026 18:28	3D Object	41 KB
Mesh_roof_corner_01_001	16.05.2026 18:28	3D Object	46 KB
Mesh_roof_top_part_01	16.05.2026 18:28	3D Object	46 KB
Mesh_wall_01_001	16.05.2026 18:28	3D Object	44 KB
Mesh_wall_corner_01_001	16.05.2026 18:28	3D Object	44 KB
Mesh_wall_door_01_001	16.05.2026 18:28	3D Object	48 KB
Mesh_wall_part_01_001	16.05.2026 18:28	3D Object	44 KB
Mesh_wall_top_01_001	16.05.2026 18:28	3D Object	46 KB
Mesh_wall_top_corner_01_001	16.05.2026 18:28	3D Object	45 KB
Mesh_wall_top_corner_02_001	16.05.2026 18:28	3D Object	45 KB
Mesh_wall_window_01_001	16.05.2026 18:28	3D Object	48 KB

Figure 14 - Export folder containing FBX files, JSON/HTML reports and Unity C# helper.

A clean role pair makes matching easier to inspect. Visible render assets and collider proxies are readable directly in the Outliner.

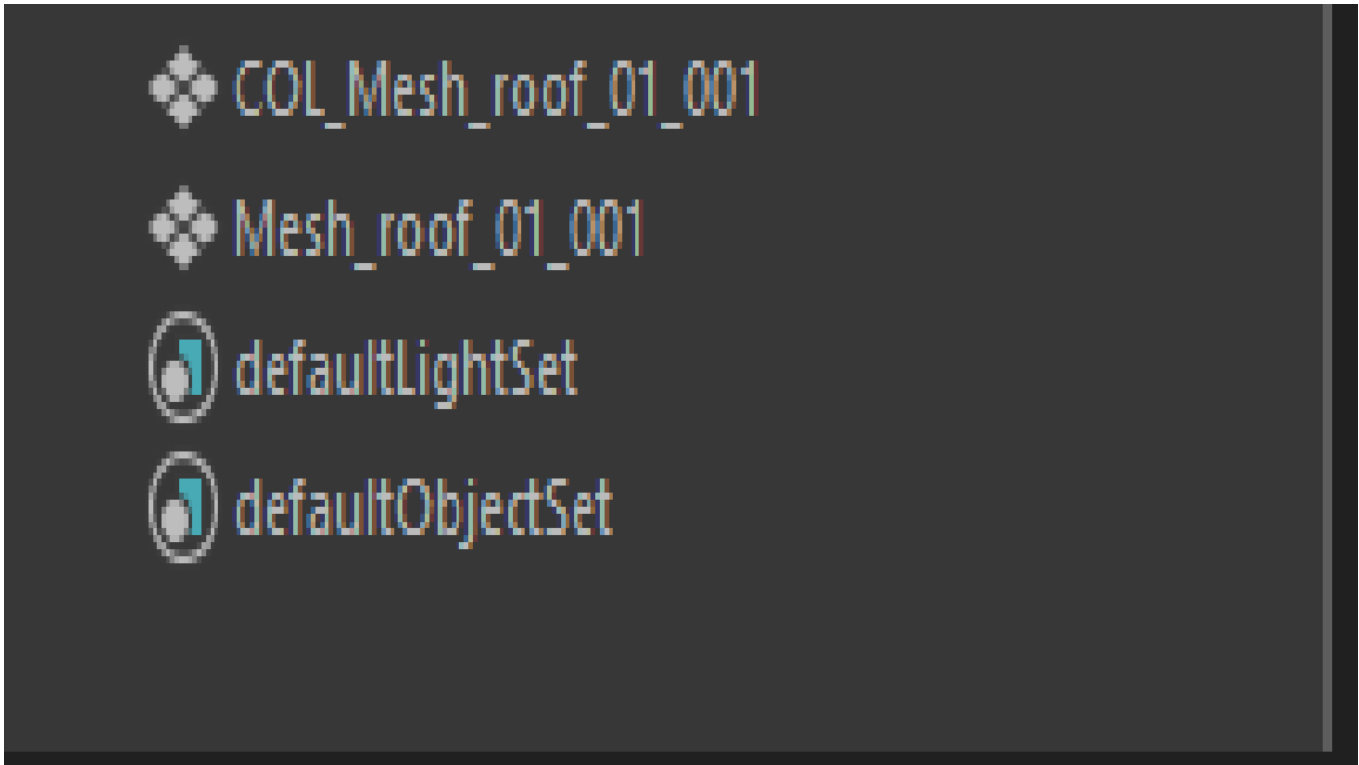


Figure 15 - Maya Outliner role pair: COL_Mesh_proxy and Mesh_render mesh.

6. Presets

Presets are starting points. Changing the dropdown alone does not modify the scene; the user must click Apply Preset to write the preset values into the interface controls.

Preset	Best for	Typical behavior
Unity Static Prop	General props, furniture, rocks, simple environment assets.	Collider proxies optional, friendly defaults, balanced validation.
Unity Environment Kit	Modular environment pieces and kit parts.	Stricter grid and pivot behavior; collider requirement and lightmap UV can be stricter.
Unity Mobile Optimized	Low-poly or mobile-focused content.	Lower polycount expectation and stricter material slot awareness.
Unity LOD / Prefab Ready	Assets that need LOD sequence and prefab readiness checks.	LOD naming/readiness receives greater emphasis.
Unity Collider Strict	Assets where every render mesh must ship with a collider or trigger proxy.	Collider Required and stricter proxy validation.

7. Interface Reference

Dashboard / Wizard tab

The Dashboard / Wizard tab is the main operator panel. It contains the preset selector, wizard flow buttons, export queue options, status cards, the asset result table and the detail panel.

Control	Function
Pipeline Preset + Apply Preset	Selects and applies a predefined rule configuration.

Analyze Selected Assets	Validates selected render mesh transforms without modifying the scene.
Apply Safe Fixes	Runs enabled cleanup actions on analyzed render meshes.
Revalidate	Runs Analyze again after changes.
Export FBX + Reports	Exports eligible assets, matched proxies and selected report outputs.
Clear Results	Clears dashboard data without modifying the Maya scene.
Reset to Defaults	Restores Unity Static Prop-style defaults without modifying the Maya scene.

Export queue options

Export queue options determine what assets and report outputs are written. The export pass performs a fresh COL/TRG_ match at export time so proxies created after the first analysis can still be included.

- Export Ready Assets: exports assets with Ready status.
- Export Warning Assets: exports assets with Warning status.
- Skip Blocked Assets: skips blocked assets by default.
- Write JSON Report: creates M2Unity_Pipeline_Report.json.
- Write HTML Report: creates M2Unity_Pipeline_Report.html.
- Write Unity AssetPostprocessor C#: creates M2Unity_ModelPostprocessor.cs.
- Triangulate FBX Export: asks the FBX exporter to triangulate output geometry.

Rules tab

Rules define what the tool checks. Each rule family focuses on a different part of asset readiness.

Rule family	Checks
Asset Scope	Front axis, ignoring COL/TRG_UCX objects as base assets, descendant mesh shape inclusion.
Naming	Enable naming rule, required render mesh prefix, naming severity, sanitize asset names and import prefix cleanup.
Geometry	Max polycount, freeze requirement, construction history and zero-thickness warnings.
Dimensions	Expected width/height/depth and tolerance rules.
Pivot	Expected pivot target, tolerance, severity and pivot fix behavior.
Grid	Grid step, bounds snap, size multiple and grid severity.
Advanced Unity Readiness	LOD, socket/locator, material slot, UV and optional lightmap UV checks.

Prep / Fix tab

Safe Fix is a conservative cleanup system. It is intentionally separate from role conversion. The tool can rename/sanitize render meshes and sync matching proxy names, but it should not turn a render mesh into COL_ or TRG_ unless the user explicitly uses a role utility button.

Safe Fix option	Behavior
Duplicate Backup Before Fix	Creates a backup duplicate before changing the asset.

Hide Backups	Hides backup duplicates after creation.
Rename / Add Prefix If Missing	Sanitizes render mesh names and adds Mesh_ or the configured render prefix.
Sync Collider Proxy Names	Keeps matching COL_/TRG_ names aligned when render mesh names are cleaned.
Freeze Transform	Freezes transforms on processed render meshes.
Delete Construction History	Deletes construction history on processed render meshes.
Move Pivot To Target	Moves pivot to the configured target without intentionally moving vertices.
Unlock Transform Attributes	Unlocks standard transform attributes if needed.
Snap Bounds Min / Pivot To Grid	Performs grid alignment based on the selected grid step.

8. Role Naming and Unity Collider Proxy Workflow

M2Unity uses explicit role names because Unity does not automatically interpret COL_ or TRG_ names in the same way Unreal interprets UCX_ names. The naming convention makes objects readable in Maya and lets the generated Unity C# postprocessor identify proxy objects after import.

```
Mesh_Table # visible render mesh
COL_Mesh_Table_01 # collider proxy mesh
COL_Mesh_Table_02 # second collider proxy part
TRG_Mesh_Table_Interaction # trigger proxy mesh
```

Naming / Role Utility buttons

Button	Result	Use case
Set Selected as Render Mesh	Mesh_<Name>	Use on visible mesh assets that should render in Unity.
Set Selected as Collider Proxy	COL_Mesh_<Name>	Use on low-poly proxy objects, including old UCX meshes.
Set Selected as Trigger Proxy	TRG_Mesh_<Name>	Use on trigger volumes or interaction zones.
Remove M2Unity Role Prefixes	Strips known role prefixes	Use when cleaning old naming before applying a new role.

Old UCX to Unity proxy conversion

Users who previously prepared Unreal-style UCX objects can convert them into Unity-style collider proxies. The role utility can strip UCX_/SM_ fragments and rename the selection into COL_Mesh_ form.

```
UCX_SM_Table_01 -> COL_Mesh_Table_01
UCX_wall_corner_01 -> COL_Mesh_wall_corner_01
```

Collider matching

Analyze and Export normalize known prefixes during comparison. The cleanest production naming remains Mesh_<AssetName>, COL_Mesh_<AssetName>_01 and TRG_Mesh_<AssetName>_Trigger, but the matching system can tolerate some older or mixed names by stripping known prefixes.

- Known role prefixes include Mesh_, COL_, TRG_, COL_Mesh_, TRG_Mesh_, UCX_ and SM_.
- Maya import / scene prefixes before known role prefixes can be interpreted and cleaned.
- Collider Required mode requires at least one matching COL_ or TRG_ proxy at Analyze and Export time.

- Collider optional mode reports and exports matched proxies when found but does not block missing proxies by default.

Export behavior with matched proxies

When exporting a render mesh, the tool refreshes matching and includes matched COL_ and TRG_ objects in the same FBX export selection. This allows a single FBX to contain the visible render mesh plus its proxy objects.

Scene objects:
 Mesh_Table
 COL_Mesh_Table_01
 COL_Mesh_Table_02
 TRG_Mesh_Table_Interaction

Exported FBX:
 Mesh_Table.fbx

FBX contains:
 Mesh_Table, COL_Mesh_Table_01, COL_Mesh_Table_02, TRG_Mesh_Table_Interaction

9. Unity AssetPostprocessor C# Helper

When enabled, the tool writes M2Unity_ModelPostprocessor.cs. This script is intended for Unity Editor use, not Maya. It should be reviewed and placed inside a Unity project under an Editor folder, such as Assets/Editor/.

Area	Generated helper behavior
Model import settings	Applies conservative importer defaults such as disabling camera/light/animation import and preserving hierarchy.
Normals and tangents	Imports normals and calculates tangents in a Unity-friendly way.
Secondary UVs	Can generate secondary UVs based on script settings.
COL_ objects	Adds MeshCollider behavior and disables renderers on collider proxy objects.
TRG_ objects	Adds MeshCollider behavior, sets isTrigger and disables renderers on trigger proxy objects.

Production note

Teams with custom Unity import pipelines should review and adapt the generated C# script before adding it to a shared project. The generated script is a helper, not a replacement for all studio import rules.

10. Reports and QA Output

M2Unity produces JSON and HTML reports. JSON is intended for technical debugging, data comparison or future pipeline integration. HTML is intended for readable artist QA and handoff.

Report	File name	Primary use
JSON	M2Unity_Pipeline_Report.json	Structured data for debugging and pipeline integration.
HTML	M2Unity_Pipeline_Report.html	Artist-friendly QA review with status cards, asset details and validation tables.
Unity helper	M2Unity_ModelPostprocessor.cs	Optional Unity Editor import helper.

Report contents

- Tool name, build and timestamp.
- Active settings and profile information.
- Export summary and per-asset export status.
- M2Unity Score, Ready/Warning/Blocked status, polycount and included collider count.
- Warnings, blocking issues, fixed actions and manual review notes.
- Collider proxy matches, target candidates and validation results.
- LOD, material, UV, socket and core validation results.

The HTML report is intended for artist-friendly QA review. It shows exported status, score, included collider count, issues and fixed actions.

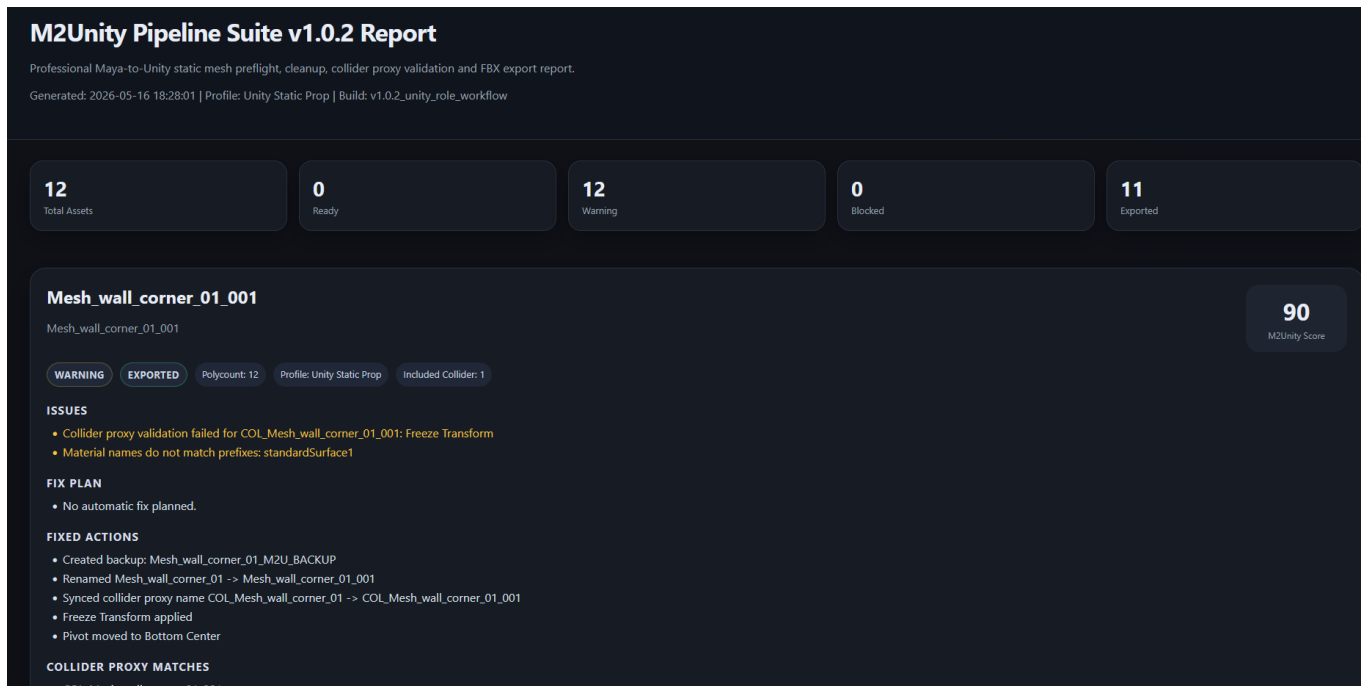


Figure 16 - HTML report summary with asset status, score, issues and fixed actions.

Core checks show pass/fail data for naming, dimensions, transforms, pivot, grid, history and other validation rules.

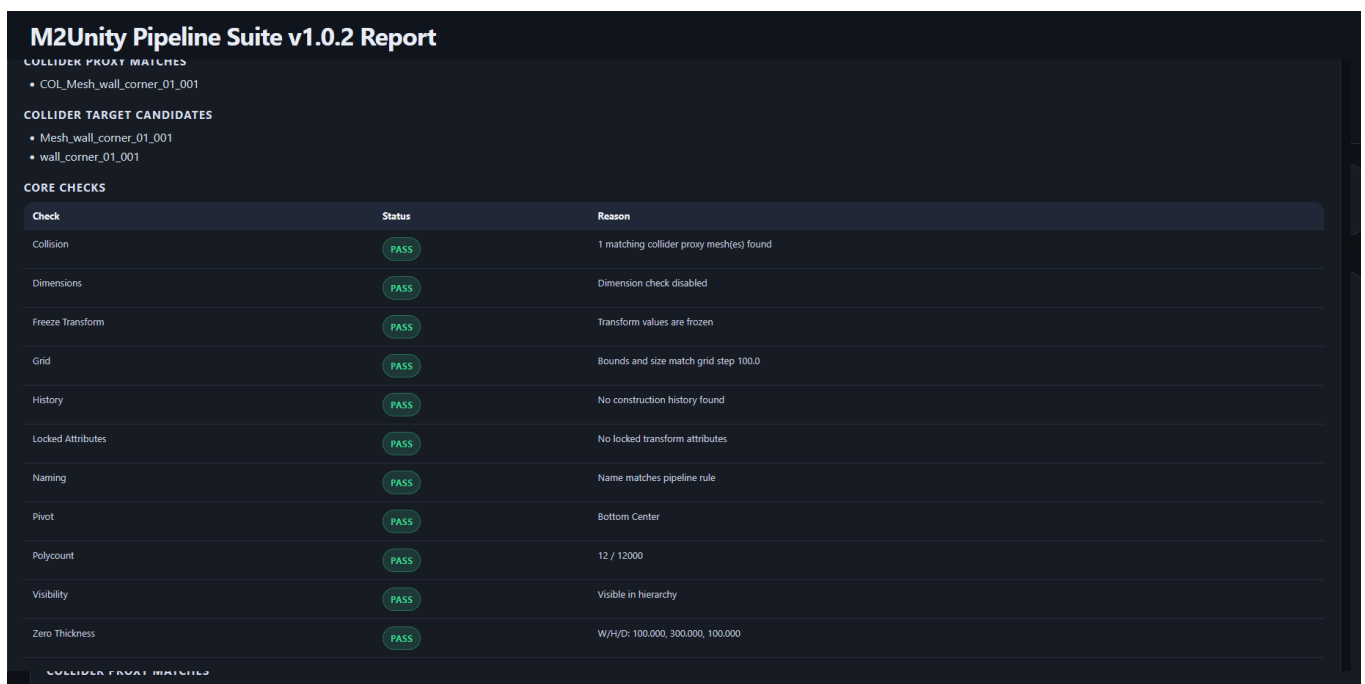


Figure 17 - HTML report core checks for analyzed assets.

Detailed tables provide structured validation results for review, troubleshooting or production handoff.

M2Unity Pipeline Suite v1.0.2 Report		
LOD / MATERIAL SLOT / UV / SOCKET READINESS		
Check	Status	Reason
LOD Group	PASS	No LOD naming detected. Single mesh workflow is valid.
Default Material	PASS	No default material detected
Material Naming	FAIL	Names without prefix: standardSurface1
Material Slot Count	PASS	1 / 8
Material Slots	PASS	1 material slot(s): standardSurface1
Lightmap UV	PASS	Lightmap UV found or not required
UV 0-1 Range	PASS	Sampled UVs are inside 0-1
UV Coordinates	PASS	25 UV coordinate(s) found
UV Overlap	INFO	Overlap calculation is advisory/manual in v1.0
Socket Locked Attrs	PASS	No locked socket transform attributes
Socket Naming	PASS	Socket names OK
Sockets	PASS	No sockets found. This is valid for simple static meshes.

Figure 18 - HTML report detailed validation table.

11. Detailed Feature Reference

This section summarizes the primary functions and options in user-facing terms. It is designed as a quick reference for artists and technical users.

Asset Scope

Option	Description
Front Axis	Defines +X, -X, +Z or -Z as the horizontal front direction used for pivot target calculations. Maya Y remains the height/up axis.
Ignore COL_/TRG_ As Base Assets	Prevents proxy meshes from being analyzed as visible render assets. Role Utility buttons can still operate on selected proxy objects.
Include Descendant Mesh Shapes	Includes mesh shapes under descendant transforms, useful for grouped assets or simple hierarchies.

Naming

Option	Description
Enable Naming Rule	Checks whether the render mesh follows the required prefix rule.
Required Prefix	Default render prefix is Mesh_. COL_, TRG_ and UCX_ are reserved for proxy roles and should not be used as render prefixes.
Sanitize Asset Names	Cleans invalid characters, namespaces and import prefixes before known role prefixes.

Geometry

Option	Description
Max Polycount	Defines the maximum allowed face count for the analyzed transform.

Freeze Required	Checks whether translate/rotate are zero and scale is one.
Clean History Required	Checks for construction history that may be unsafe or unnecessary for export.
Zero Thickness Warning	Detects accidental planes, collapsed objects or near-zero dimensions.

Dimensions, Pivot and Grid

Option	Description
Dimension Check	Compares bounding-box width, height and depth against expected values. Safe Fix does not resize geometry automatically.
Pivot Target	Defines expected pivot position such as Bottom Center, Bottom Front Center or Top Center.
Grid Step	Defines the modular grid interval. A Unity-oriented default commonly represents one-meter modular alignment in a Maya-centimeter workflow.

Advanced Unity Readiness

Option	Description
LOD Readiness	Checks Unity-friendly names such as Mesh_Rock_LOD0, Mesh_Rock_LOD1 and Mesh_Rock_LOD2.
Socket / Locator Readiness	Checks optional helper names such as SOCKET_Grip, SOCKET_Attach or SOCKET_Effect.
Material Slot Readiness	Reports missing assignments, excessive material slots or names that do not match accepted material prefixes.
UV Readiness	Checks UV set existence and conservative UV range behavior. It is not a full UV overlap solver.
Lightmap UV Requirement	Checks for configured secondary/lightmap UV names when required by the selected workflow.

Collider Proxy

Option	Description
Collider Requirement	Off, Collider required, Collider optional or No custom collider allowed.
Match Mode	Base asset name or Exact custom target name.
Accept Multiple Collider Proxy Parts	Allows multiple COL/TRG_ proxy parts per render mesh.
Validate Matched Collider Proxy Meshes	Checks proxy face count, missing shapes, locked attributes and other proxy readiness concerns.
Max Collider Faces	Default conservative proxy limit is 255 faces, useful when proxy meshes may be used in convex-style workflows.

12. Recommended Production Workflows

Workflow	Steps
----------	-------

Simple Static Prop	Select visible mesh -> Set Selected as Render Mesh if needed -> Unity Static Prop preset -> Analyze -> Safe Fix -> Revalidate -> Export.
Static Prop With Collider Proxy	Set render mesh role, set collider proxy role, select render mesh, enable Collider Required or use Unity Collider Strict, Analyze, confirm match, Export.
Old Unreal UCX To Unity Collider Proxy	Select old UCX object -> Set Selected as Collider Proxy -> select Mesh_render asset -> Analyze -> Export.
Trigger Proxy Setup	Select trigger volume mesh -> Set Selected as Trigger Proxy -> export with matched render asset -> use generated Unity postprocessor to create trigger MeshCollider behavior.
Modular Environment Kit	Use strict pivot/grid rules, consistent Mesh_naming, simple COL_Mesh_proxies and optional lightmap UV checks.

Recommended naming examples

```
Mesh_Chair
Mesh_Table
Mesh_Rock_LOD0
COL_Mesh_Table_01
COL_Mesh_Table_02
TRG_Mesh_Door_Interaction
```

13. Safety Rules, Limitations and Troubleshooting

Safe Fix safety rules

- Keep Duplicate Backup Before Fix enabled when working on important scenes.
- Review the analysis result before applying fixes.
- Safe Fix does not rebuild topology, resize geometry or automatically create complex collider geometry.
- Safe Fix should not convert render meshes into COL_ or TRG_ role meshes; role conversion is explicit.
- Revalidate after fixes and review the HTML report after export.

Known limitations

- The tool does not open Unity or import FBX files into Unity from Maya.
- Generated C# scripts must be manually copied into a Unity Editor folder and reviewed.
- COL_ and TRG_ are M2Unity conventions, not built-in Unity naming rules.
- UV overlap detection is conservative and does not replace specialist UV QA.
- Collider performance and final Unity behavior must still be reviewed by the user.
- Studio-specific import settings may require custom edits to the generated C# postprocessor.

Troubleshooting

Problem	Check or fix
No assets are analyzed.	Select visible Mesh_render assets. COL_ and TRG_proxy objects are ignored as base assets by default.
Collider Required blocks the asset.	Confirm that a matching COL_ or TRG_proxy exists, for example Mesh_Table and COL_Mesh_Table_01.
Old UCX object is not treated as Unity proxy.	Select the old UCX mesh and click Set Selected as Collider Proxy.

Unity does not convert COL_ objects to colliders.	Copy M2Unity_ModelPostprocessor.cs into an Assets/Editor folder and reimport/review the FBX.
Shelf button is missing.	Run reinstall_m2unity_pipeline_suite_shelf_button.py.
FBX export fails.	Confirm the Maya FBX plug-in is available and the export folder is writable.

14. Licence and Usage Terms Summary

This section summarizes the provided licence terms. The full licence text should remain with the public release package.

Area	Summary
Ownership	M2Unity Pipeline Suite and related files remain the intellectual property of the author.
Allowed use	Personal, educational, portfolio, evaluation, internal and commercial production workflow use is allowed.
Commercial production	Users may use the tool to create or deliver their own commercial assets and Unity projects.
Restrictions	Users may not resell, redistribute, re-upload, rebrand, bundle or claim ownership of the tool.
Generated outputs	FBX exports, JSON/HTML reports and generated Unity helper scripts may be used in the user's own workflow.
No warranty	The tool is provided as is; users are responsible for backups, testing and reviewing generated output.

Required public credit
Developer: Sonat Birdane

15. Appendix A - Full Licence Terms

M2Unity Pipeline Suite — LICENCE TERMS

Developer: Sonat Birdane

IMPORTANT Please read these terms before downloading, installing, using, sharing, modifying, or redistributing this tool.

M2Unity Pipeline Suite and all related files, source code, scripts, installer files, shelf setup files, icons, documentation, examples, generated report templates and packaged releases remain the intellectual property of the author.

The tool is provided for use under the terms described in this document.

These terms do not transfer ownership of the tool, source code, branding, icons, documentation, or related materials to the user.

You may:

Download and use this tool for personal, educational, portfolio, evaluation and internal workflow use.

Use this tool in both non-commercial and commercial content production workflows.

Install and run the tool on your own system.

Use the tool to prepare, validate, fix, export and document your own Maya static mesh assets.

Use the tool while creating assets, games, projects, marketplace content, client work or studio work.

Use exported FBX files, generated reports and generated Unity helper scripts as part of your own production workflow.

Privately modify the tool for your own personal testing or internal workflow needs, as long as the modified version is not publicly shared, redistributed, sold, or represented as an official release.

You are allowed to use M2Unity Pipeline Suite as part of your own commercial production workflow.

This means you may use the tool to create, prepare, validate, export and deliver your own commercial 3D assets, games, Unity projects, marketplace products, client work, studio work or internal production content.

However, commercial production use does not grant the right to sell, redistribute, repack, sublicense, bundle, publish, or include the tool itself as part of another paid product, service, asset pack, training package, software bundle, or marketplace listing without explicit written permission from the author.

M2Unity Pipeline Suite may be distributed for free through official release locations controlled by the author.

The author may also provide an optional pay-what-you-want or donation-based download through official platforms.

Optional payment or support does not grant resale, redistribution, sublicensing, ownership, or rebranding rights unless a separate written agreement is provided by the author.

You may not:

Re-upload this tool to other websites, drives, forums, repositories, marketplaces, file-sharing platforms, or third-party distribution channels without permission.

Redistribute the files in modified or unmodified form without explicit permission from the author.

Claim authorship or ownership of the tool.

Sell, sublicense, rent, lease, bundle, package, or commercially redistribute the tool itself.

Include the tool itself as part of a paid product, paid service, asset pack, software bundle, course package, commercial distribution, or marketplace listing without explicit written permission from the author.

Publish edited, modified, repackaged, renamed, or rebranded versions as if they were official releases.

Remove, hide, or alter author credits, copyright notices, licence terms, documentation notices, or distribution information.

Use the tool name, branding, icon, or documentation in a misleading way that suggests official endorsement, partnership, or authorship.

Use generated installer or shelf files to misrepresent the origin of the tool.

Use the tool in a way that violates Autodesk Maya, Unity, or any third-party software licence terms.

This tool should only be downloaded from official release locations controlled by the author.

If you received the files from an unofficial source, you should obtain the latest official version from an official distribution location instead.

Unofficial copies may be outdated, modified, unsafe, incomplete, or unsupported.

Private modification for personal testing, learning, or internal workflow adaptation is allowed.

However, modified versions may not be:

Publicly shared

Redistributed

Sold

Uploaded to another platform

Represented as official

Used to mislead users about the original author or official release status

Bundled into another product without explicit written permission

If you want to publish, fork, redistribute, translate, bundle, or commercially modify the tool, you must obtain explicit permission from the author unless the author provides a separate open-source licence stating otherwise.

The tool may generate files such as:

FBX exports JSON reports HTML reports Unity AssetPostprocessor C# helper scripts

You may use generated output files in your own personal, educational, internal, or commercial production workflows.

Generated reports and helper scripts are provided for workflow support and review. They do not remove the need for manual QA, technical review, or project-specific import validation.

If you share generated reports publicly, you are responsible for removing any private project names, asset names, file paths, client names, or other confidential information that may appear in those reports.

This tool is provided “as is” without warranty of any kind.

The author does not guarantee that the tool will be error-free, uninterrupted, compatible with every Maya version, compatible with every Unity version, compatible with every studio environment, or suitable for every production pipeline.

The author is not responsible for:

Data loss

Scene corruption

Export issues

Import issues

Broken files

Workflow interruption

Production delays

Compatibility issues

Unsupported Maya versions

Unsupported Unity versions

Third-party plug-in conflicts

Incorrect user configuration

Problems caused by unofficial, modified, or redistributed versions

Problems caused by manually edited generated scripts

Problems caused by adding generated scripts to Unity projects without review

Always keep backups of important Maya scenes, Unity projects and production files before using automated cleanup, validation, import, export, or postprocessing tools.

Users are responsible for reviewing all changes made by the tool.

Before using the tool in production, users should:

Test it on duplicate scenes.

Keep backups of important files.

Review Safe Fix settings before applying them.

Review generated FBX files before delivery.

Review generated JSON and HTML reports.

Review generated Unity C# scripts before placing them inside a Unity project.

Confirm that generated collider and trigger behavior matches project requirements.

Confirm that the tool fits the intended production pipeline.

M2Unity Pipeline Suite is designed to work with Autodesk Maya and Unity workflows.

Autodesk Maya, Unity, and related names, trademarks, technologies and software products belong to their respective owners.

This tool is not officially affiliated with, endorsed by, sponsored by, or maintained by Autodesk, Unity Technologies, or any other third-party software vendor unless explicitly stated by those companies.

Users must follow the licence terms of all third-party software used with this tool.

The author may provide updates, fixes, documentation improvements, or support at the author's discretion.

There is no guarantee of ongoing maintenance, compatibility updates, custom feature development, or response time unless a separate written agreement exists.

Future versions may change features, file names, installer behavior, documentation, UI layout, naming conventions, or generated script behavior.

To the maximum extent permitted by applicable law, the author shall not be liable for any direct, indirect, incidental, special, consequential, exemplary, production-related, business-related, or financial damages arising from the use or inability to use this tool.

This includes but is not limited to:

Lost work

Lost time

Lost revenue

Production delays

Missed deadlines

Asset corruption

Project corruption

Client delivery problems

Pipeline disruption

Incorrect export or import results

Use the tool at your own risk.

By downloading, installing, running, modifying, or using M2Unity Pipeline Suite, you agree to these terms.

If you do not agree with these terms, do not use the tool.

You may use M2Unity Pipeline Suite for personal, educational, portfolio, internal and commercial production work.

You may not resell, redistribute, re-upload, rebrand, bundle, or claim ownership of the tool.

You are responsible for backups, testing, reviewing generated output and confirming that the tool fits your workflow.

The tool is provided as is, with no warranty.

16. Appendix B - File and Report Names

```
Main tool: m2unity_pipeline_suite_v1_0.py
Launcher: M2Unity Pipeline Suite.py
Installer: M2Unity_Pipeline_Suite_v1_0_Easy_Installer.py
Shelf reinstall: reinstall_m2unity_pipeline_suite_shelf_button.py
Icon: m2unity_pipeline_suite_icon.png
JSON report: M2Unity_Pipeline_Report.json
HTML report: M2Unity_Pipeline_Report.html
Unity helper: M2Unity_ModelPostprocessor.cs
```

17. Appendix C - Documentation Source Notes

This PDF was prepared from the v1.0.2 README, FUNCTIONS guide, LICENCE/TERMS file and the provided UI screenshots. Example paths are generic and are not required installation locations.

No private machine path or private personal data is included. The developer credit used throughout the documentation is Developer: Sonat Birdane.